Pepper/Chili Disease Detection Using Transfer Learning

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1. Motivation & Impact

Problem: Indian agriculture loses 50k-90k Cr/year to misdiagnosed diseases. Bacterial Spot destroys **20–30%** of pepper yields.
Solution: Highly accurate, lightweight CNN deployed as an accessible web app for early intervention.

2. Dataset & Augmentation

Source: PlantVillage (*Capsicum annuum*)
Size: 2,475 images (Healthy vs. Bacterial Spot)
Splits: 70% Train, 15% Val, 15% Test

Augmentations applied: RandomCrop (224×224), Horizontal/Vertical Flips, ColorJitter, RandomRotation ($\pm 15^\circ$).

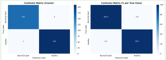
3. Model Architecture

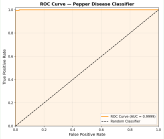
Model: EfficientNet-B0 (`timm`) ~4.01M params.
Config: Custom `Linear(1280` $\rightarrow$ `2)` head.
Training: AdamW, CosineAnnealingLR. Trained for 10 epochs (~15 mins on Kaggle T4 GPU).

4. Evaluation Results

Achieved near-perfect classification on unseen test data (Epoch 10): **Test Accuracy: 99.73%** — **ROC-AUC: 0.9999**

Class	Precision	Recall	Support
Bacterial Spot	99.38%	100.00%	161
Healthy	100.00%	99.53%	211





5. Ablation Study

Evaluated at Epoch 3 to test pretraining impact:

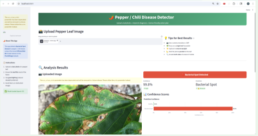
- **ImageNet Pretrained:** 99.46% Accuracy
- **Random Initialization:** 49.19% Accuracy

Conclusion: Transfer learning yields a **+50.00%** early gain and vastly superior feature extraction.

6. Live Web Application

Deployed globally via **Hugging Face Spaces**.

- **Diagnosis:** Instant inference & confidence.
- **Action Plans:** Farmer-friendly treatments.



7. Conclusion & Future Work

Lightweight architectures with transfer learning achieve **>99.5% reliability**, proving viability for edge-devices.
Future: Grad-CAM integration and multi-class.

8. Project Links

Live App: HuggingFace Spaces — **Code:** GitHub Repo